

GLOSSARY

ACID

A substance that when dissolved in water produces positively charged ions of hydrogen. Lemon juice and vinegar are examples of acids.

AIR RESISTANCE

A force that acts on an object moving through air, in the opposite direction to the object's motion.

ATOM

The smallest part of an element.

ATTRACTION

A force that pulls things together.

BACTERIA

Microscopic, single-celled organisms. Some can cause illness, but most are harmless.

BASE

A kind of substance that reacts with an acid to give water and salt.

BOND

A force that holds together tiny particles such as atoms and molecules.

CARBON DIOXIDE

A compound found as a gas in the air all around us. We breathe out carbon dioxide as a waste product.

CELL

1. The smallest living part of a living thing. Plants and animals are made of up to trillions of cells. 2. A chemical device that forms part of an electric battery.

CELLULOSE

A compound that forms the tough fibers of plant cell walls.

CHEMICAL

A compound or element that can change when combined with another substance. Chemicals can be liquids, solids, or gases.

CIRCUIT

A complete and closed path, around which an electric current can flow.

COMPOUND

A chemical made of two or more elements. For example, water is a compound, made of the elements hydrogen and oxygen.

COMPRESSION

A squashing force, such as that experienced by weight-bearing materials in buildings.

CONDUCTOR

A substance through which heat or electricity can easily pass.

CRYSTAL

A regular arrangement of atoms or molecules held together by bonds in a solid.

DENSITY

The amount of mass there is in a particular volume.

DNA

Short for deoxyribonucleic acid. DNA is a compound found in the cells of all living things. DNA holds coded instructions (genes) that control what people, animals, and plants look like and how they function.

ELECTRIC CURRENT

Movement of electric charge.

ELECTRON

A tiny particle in an atom that has a negative electric charge.

ELEMENT

A substance made of just one type of atom that cannot be broken down into a simpler substance by chemical reactions.

ENERGY

The ability to make things happen. Energy has different forms, such as electrical energy and kinetic energy (movement).

EVAPORATION

The process by which a liquid turns into a gas, usually because of an increase in temperature.

FILTRATION

The process of separating solids from liquids by passing the mixture through a filter.

FOSSIL

The remains or traces of a long-dead animal or plant that have been preserved in rock.

GENE

Part of the DNA found in every living cell. Genes help to make living things the way they are, such as a tall, brown-eyed person or a leafy plant.

GENOME

The complete set of information carried by all the genes in a living thing. The human genome consists of about 20,000 genes.

GLUCOSE

A compound made by plants during photosynthesis. Glucose is a sugar, and is used for energy.

GRAVITY

An attractive force between two objects. Gravity keeps you on the ground instead of floating around.

HELIX

A shape that winds around like a spiral staircase. DNA molecules have a double helix shape.

INSULATOR

A substance through which heat or electricity cannot pass easily.

ION

An atom that has gained or lost electrons, so gaining a negative or positive electric charge.